

10. A technician prepared 250 cm<sup>3</sup> of a 0.50 mol/dm<sup>3</sup> solution of citric acid, C<sub>6</sub>H<sub>8</sub>O<sub>7</sub>.

- (a) Calculate the number of moles of citric acid in the solution. [2]

$$\text{concentration (mol/dm}^3\text{)} = \frac{\text{number of moles}}{\text{volume (dm}^3\text{)}}$$

$$1 \text{ dm}^3 = 1000 \text{ cm}^3$$

Number of moles = ..... mol

- (b) Calculate the mass of citric acid in the solution. [2]

$$A_r(\text{H}) = 1$$

$$A_r(\text{C}) = 12$$

$$A_r(\text{O}) = 16$$

$$\text{number of moles} = \frac{\text{mass (g)}}{M_r}$$

Mass = ..... g

**END OF PAPER**

